PHYS 4041/5041
Fall 2023
Homework 2
Due: In class, Friday September 15, 2023

1. Find the general solution $I(t)$ for the sinusoidally-driven RL circuit:

$$L\frac{dI}{dt} + RI = V_0 \cos(\omega t),$$

where L, R, V_0, and ω are constants.

2. Find the solution to

$$y'' - 3y' + 2y = 0,$$

subject to $y(0) = 0$ and $y'(0) = 1$.

3. Find the general solutions to the following (note $'$ means d/dx):
   (a) $y'' + y = 2xe^x$
   (b) $y'' + 2y' = 2x$

4. Evaluate the following:
   (a) $\int_{-1}^{1} \cos(\pi x)\, \delta(\pi x)\, dx$
   (b) $\int_{-1}^{1} \cosh(x)\, \delta''(2x-1)\, dx$
   (c) $\int_{-1}^{1} \cos x\, \delta(x-2)\, dx$

5. Determine whether or not the the following series converge. If it converges, is the convergence absolute or conditional?
   (a) $\sum_{n=0}^{\infty} \frac{n}{\sqrt{n^2+1}}$
   (b) $\sum_{n=0}^{\infty} \frac{4^{2n}}{3^{3n}}$
   (c) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$

6. The Langevin theory of paramagnetism leads to the following expression for the magnetization M:

$$M(x) = n\mu \left(\frac{\cosh x}{\sinh x} - \frac{1}{x} \right),$$

where n is the atomic number density, μ is the atomic magnetic moment, $x \equiv \mu B/(k_B T)$, B is the applied magnetic field, k_B is Boltzmann 's constant, and T is the temperature. Determine the first two non-zero terms in the power series for $M(x)$ (i.e., for low B or high T).

7. Planck's theory of quantized oscillators leads to an average energy of the oscillators $\langle E \rangle$ given by:

$$\langle E \rangle = \frac{\sum_{n=1}^{n=\infty} nE_0 \exp(-nE_0/k_BT)}{\sum_{n=0}^{n=\infty} \exp(-nE_0/k_BT)},$$

where E_0 is a constant energy, k_B is Boltzmann 's constant, and T is the temperature.

(a) Evaluate the sum in the denominator by noting that it is a geometric series.

(b) Evaluate the sum in the numerator by noting that it is closely related to $d/d\beta$ operating the denominator, where $\beta = -1/(k_BT)$. Your result for $\langle E \rangle$ should be

$$\langle E \rangle = \frac{E_0}{\exp(E_0/k_BT) - 1}.$$

(c) Show that this formula reduces to the classical result, $\langle E \rangle = k_BT$, for $k_BT \gg E_0$.

8. Solve

$$y'' + 4y' + 5y = \delta(t - t_0)$$

for $t \geq 0$, subject to $y(0) = y'(0) = 0$, and assuming $t_0 > 0$.

9. Consider the differential equation

$$\frac{d^2y}{dt^2} + k\frac{dy}{dt} = f(t),$$

with $y(0) = \dot{y}(0) = 0$ and $k > 0$.

(a) Find the Green's function for this problem that will be useful for finding the solution for $t > 0$.

(b) Find the solution for $t > 0$ for

$$f(t) = f_0 \exp(-\lambda t),$$

where f_0 and $\lambda > 0$ are constants.